

7.5.1 – 7.5.3 Conics Worksheet

1. Find the vertex, focus, and directrix of the graph $y = 3x^2 - 12x + 7$.
2. Suppose that a parabola has a vertex of $(-2, 4)$ and a directrix of $x = 1$.
 - a. State the focus of this parabola
 - b. Find the equation for the parabola
 - c. *Carefully* sketch a picture of this parabola. Label the vertex, directrix, focus, and plot at least 3 points.
3. Consider the equation of a conic: $x^2 + 2x + y^2 + 8y - 8 = 0$.
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. If this equation represents a parabola state its focus. If this equation represents an ellipse or hyperbola state its foci. If it is a circle state its center.
 - c. If this equation represents a parabola state its directrix. If this equation represents an ellipse or hyperbola state its vertices. If it is a circle state its radius.
 - d. Carefully sketch a graph of the above equation.
4. Consider the equation of a conic: $-x^2 - 8y - 4 = 4y^2 - 4x$
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. If this equation represents a parabola state its focus. If this equation represents an ellipse or hyperbola state its foci. If it is a circle state its center.
 - c. If this equation represents a parabola state its directrix. If this equation represents an ellipse or hyperbola state its vertices. If it is a circle state its radius.
 - d. Carefully sketch a graph of the above equation.
5. Consider the equation of a conic: $x^2 - 4y^2 - 2x + 8y - 7 = 0$
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. If this equation represents a parabola state its focus. If this equation represents an ellipse or hyperbola state its foci. If it is a circle state its center.
 - c. If this equation represents a parabola state its directrix. If this equation represents an ellipse or hyperbola state its vertices. If it is a circle state its radius.
 - d. Carefully sketch a graph of the above equation.
6. Find an equation for the hyperbola with center $(3, 3)$, one focus at $(3, 8)$, and a vertex at $(3, 0)$.
7. Consider the equation of a conic: $x^2 = 6y - y^2$
 - a. Does this equation represent a parabola, ellipse, circle, or hyperbola?
 - b. If this equation represents a parabola state its focus. If this equation represents an ellipse or hyperbola state its foci. If it is a circle state its center.
 - c. If this equation represents a parabola state its directrix. If this equation represents an ellipse or hyperbola state its vertices. If it is a circle state its radius.
 - d. Carefully sketch a graph of the above equation.

8. Consider the equation: $y^2 - 8x - 8y - 8 = 0$
- Does this equation represent a parabola, ellipse, or hyperbola?
 - If this equation represents a parabola state its focus. If this equation represents an ellipse or hyperbola state its foci.
 - If this equation represents a parabola state its directrix. If this equation represents an ellipse or hyperbola state its vertices.
 - Carefully sketch a graph of the above equation.
9. Consider the conic given by the equation: $-x^2 + 2x + 4y^2 + 8y - 8 = 0$.
- Does this equation represent a parabola, ellipse, or hyperbola?
 - If this equation represents a parabola state its focus. If this equation represents an ellipse or hyperbola state its foci. If it is a circle state its center.
 - If this equation represents a parabola state its directrix. If this equation represents an ellipse or hyperbola state its vertices. If it is a circle state its radius.
 - Carefully sketch a graph of the above equation.